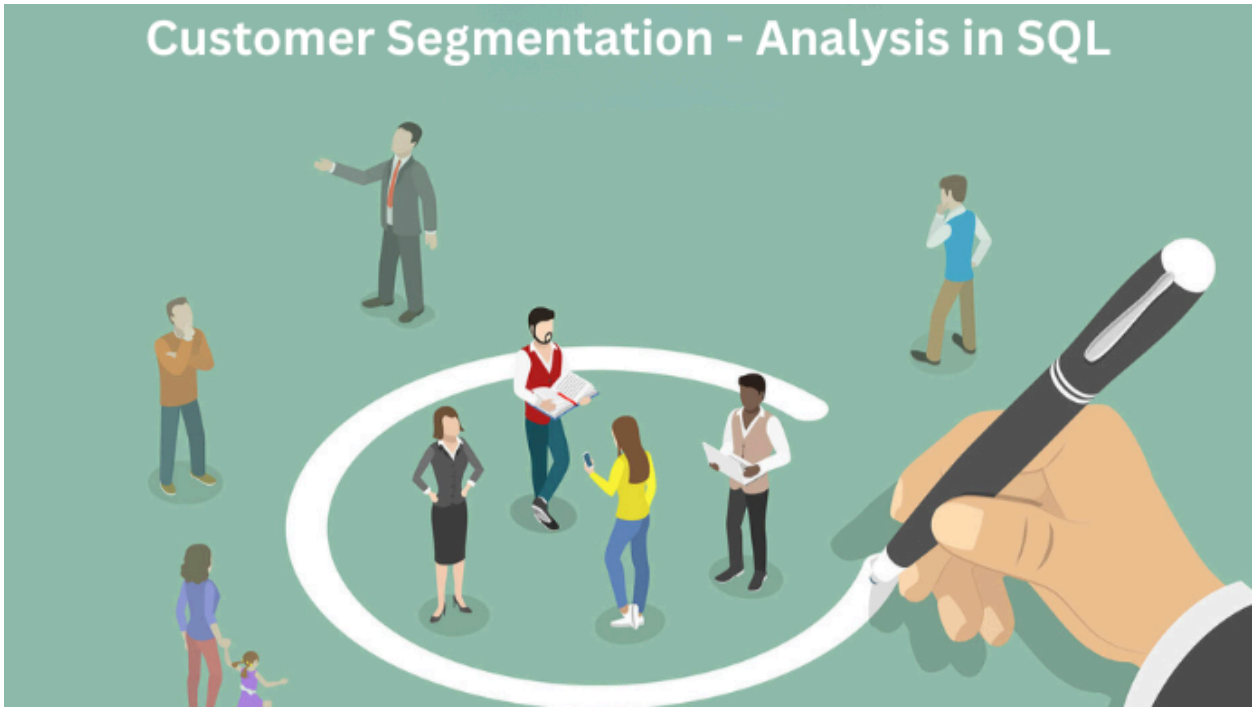


Customer Segmentation Using RFM Analysis

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Course: SQL and Databases

Project Overview

This project uses the RFM (Recency, Frequency, Monetary) framework to segment customers from the SNOWFLAKE_SAMPLE_DATA.TPCH_SF1 dataset.

The goal is to identify high-value customers, understand ordering behavior, and generate actionable insights to guide marketing strategies.

Methodology

Data Source

SNOWFLAKE_SAMPLE_DATA.TPCH_SF1

Tables Used

- CUSTOMER
- ORDERS
- LINEITEM
- NATION

Approach Summary

1. Extract customer, order, and revenue information.
2. Calculate R (Recency), F (Frequency), and M (Monetary) metrics per customer.
3. Assign quartile-based RFM scores (1–5) for each metric.
4. Combine scores into an overall RFM Score (R+F+M).
5. Create customer segments (e.g., “Platinum”, “Gold”, “Silver”, “Bronze”) based on score combinations.

Tools & Environment

- **Platform:** Snowflake (Using Sample Data TPCH_SF1)
- **Language:** SQL
- **Analysis Type:** RFM Segmentation and Scoring
- **Output:** Customer Segments and Insights Report

Part 1 – Customer Segmentation (RFM)

Objective

Segment customers based on Recency, Frequency, and Monetary value (RFM).

The purpose is to identify high-value, loyal, and at-risk customers to help the marketing team tailor campaigns that drive engagement and retention.

SQL Query

```
USE SCHEMA SNOWFLAKE_SAMPLE_DATA.TPCH_SF1;
-- ============================================================================
-- 1. Customer base with nation info (join customer → nation to add
country details)
-- ============================================================================
WITH customer_base AS (
    SELECT
        c.c_custkey AS customer_id,
        c.c_name AS customer_name,
        c.c_address AS customer_address,
        c.c_phone AS customer_phone,
        c.c_acctbal AS account_balance,
        c.c_mktsegment AS market_segment,
        n.n_name AS nation_name
    FROM
        customer c
        INNER JOIN nation n ON c.c_nationkey = n.n_nationkey -- match
customer to their nation
),
-- ============================================================================
-- 2. Orders and line items
-- ============================================================================
orders_base AS (
    SELECT
        o.o_orderkey AS order_id,
        o.o_custkey AS customer_id,
        o.o_orderdate AS order_date
    FROM
        orders o
),
```

```
lineitem_base AS (
    SELECT
        l.l_orderkey AS order_id,
        l.l_extendedprice AS extended_price,
        l.l_discount AS discount
    FROM
        lineitem l
),
-- ============================================================================
-- 3. Snapshot date used for recency calculation (latest order in dataset)
-- ============================================================================
snapshot_cte AS (
    SELECT
        MAX(order_date) AS snapshot_date
    FROM
        orders_base
),
-- ============================================================================
-- 4. Aggregate customer-level RFM metrics
--    Calculate frequency, monetary value, average order value, and order
dates
-- ============================================================================
customer_order_summary AS (
    SELECT
        c.customer_id,
        -- Frequency: total number of orders per customer
        COUNT(DISTINCT o.order_id) AS total_orders,
        -- Monetary: total revenue per customer
        ROUND(
            COALESCE(
                SUM(l.extended_price * (1 - l.discount)),
                0
            ),
            2
        ) AS total_revenue,
        -- Average Order Value: revenue / number of orders
        ROUND(
            COALESCE(
                NULLIF(SUM(l.extended_price * (1 - l.discount)), 0) /
NULLIF(COUNT(DISTINCT o.order_id), 0),
                0
            ),
            2
```

```
        ) AS avg_order_value,
        -- First and last order dates for recency calculation
        MIN(o.order_date) AS first_order_date,
        MAX(o.order_date) AS last_order_date
    FROM
        customer_base c -- Join orders for each customer (may be no orders,
hence LEFT JOIN)
        LEFT JOIN orders_base o ON c.customer_id = o.customer_id -- Join
line items for each order to calculate revenue and AOV
        LEFT JOIN lineitem_base l ON o.order_id = l.order_id -- Group by
customer to aggregate all metrics at customer level
    GROUP BY
        c.customer_id
),
-- ============================================================================
-- 5. Recency calculation
--    Calculate days since the last order for each customer
--    If the customer has no orders, use a default early date
('1900-01-01')
-- ============================================================================
recency_calc AS (
    SELECT
        s.customer_id,
        s.total_orders,
        s.total_revenue,
        s.avg_order_value,
        s.first_order_date,
        s.last_order_date,
        -- Recency in days: difference between snapshot date and last order
date
        DATEDIFF(
            DAY,
            COALESCE(s.last_order_date, '1900-01-01'),
            snap.snapshot_date
        ) AS recency_days
    FROM
        customer_order_summary s
        CROSS JOIN snapshot_cte snap
),
-- ============================================================================
-- 6. RFM scoring using NTILE
--    Assign scores (1-5) for Recency, Frequency, and Monetary metrics
--    Higher score = better customer
```

```
-- ============================================================================
rfm_scores AS (
    SELECT
        r.customer_id,
        r.recency_days,
        r.total_orders,
        r.total_revenue,
        r.avg_order_value,
        r.first_order_date,
        r.last_order_date,
        -- Recency score: lower recency_days = more recent = better
        -- DESC used so most recent customers get highest score (5)
        NTILE(5) OVER (
            ORDER BY
                r.recency_days DESC
        ) AS r_score,
        -- Frequency score: higher total_orders = better
        -- ASC used so customers with the most orders get the highest score
(5)
        NTILE(5) OVER (
            ORDER BY
                COALESCE(r.total_orders, 0) ASC
        ) AS f_score,
        -- Monetary score: higher total_revenue = better
        -- ASC used so customers with the highest revenue get the highest
score (5)
        NTILE(5) OVER (
            ORDER BY
                COALESCE(r.total_revenue, 0) ASC
        ) AS m_score,
        -- Combined RFM score: sum of Recency, Frequency, and Monetary
scores
        (r_score + f_score + m_score) AS rfm_score
    FROM
        recency_calc r
),
-- ============================================================================
-- 7. Final enriched RFM dataset
--    Combine RFM scores with customer details and assign RFM segments
-- ============================================================================
final_rfm AS (
    SELECT
        r.customer_id,
```

```
        cb.customer_name,
        cb.customer_address,
        cb.customer_phone,
        cb.account_balance,
        cb.market_segment,
        cb.nation_name,
        -- Last order date and aggregated metrics
        r.last_order_date,
        r.total_revenue,
        r.total_orders,
        r.avg_order_value,
        r.recency_days,
        -- RFM scores
        r.r_score,
        r.f_score,
        r.m_score,
        r.rfm_score,
        -- Segment assignment based on combined RFM score
        CASE
            WHEN r.rfm_score BETWEEN 12 AND 15 THEN 'Platinum (High Value)'
            WHEN r.rfm_score BETWEEN 8 AND 11 THEN 'Gold (Loyal)'
            WHEN r.rfm_score BETWEEN 4 AND 7 THEN 'Silver (At Risk)'
            ELSE 'Bronze (Inactive)'
        END AS rfm_segment
    FROM
        rfm_scores r
        LEFT JOIN customer_base cb ON r.customer_id = cb.customer_id --
Join to get customer details
)
-- ============================================================================
-- Final output: top 10 customers by RFM ranking
-- ============================================================================
SELECT
    *
FROM
    final_rfm
ORDER BY
    rfm_score DESC,      -- highest RFM score first
    total_revenue DESC, -- higher spenders next
    recency_days ASC,    -- more recent customers next
    total_orders DESC    -- most frequent buyers last
LIMIT
    10;
```

Output Screenshot (Top 10 Rows)

Results (just now)

Table | Chart — 10 rows — 77ms

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_ADDRESS	CUSTOMER_PHONE	ACCOUNT_BALANCE	MARKET_SEGMENT	NATION_NAME	LAST_ORDER_DATE	TOTAL_R
1	95257	Customer#000095257	aYNvFHtnZ3nfM3KhZSS6aIZbWfj407VfjEyQ8O	12-704-221-9665	7192.11	HOUSEHOLD	BRAZIL	1998-06-16	625
2	131113	Customer#000131113	Kfg65w3gkITu44xn15QOLy	15-698-563-4092	679.25	BUILDING	ETHIOPIA	1998-08-02	601
3	134380	Customer#000134380	j0J3qLKeB6clpFxNDB46lDrZyYzT	10-184-851-8092	3599.31	FURNITURE	ALGERIA	1998-07-08	607
4	20194	Customer#000020194	HWUhzU D4OeqKW5qrcvb7tKTH4N4RDE	33-506-798-6469	1327.63	AUTOMOBILE	UNITED KINGDOM	1998-07-27	584
5	148750	Customer#000148750	4DOJ6YXrSlGexdU,Hhhd1ET,	28-219-585-6544	474.89	AUTOMOBILE	CHINA	1998-07-22	584
6	28180	Customer#000028180	l1Eo8l,pXZYXAeE	16-685-528-9933	3391.46	AUTOMOBILE	FRANCE	1998-08-02	583
7	35398	Customer#000035398	yHMMJpM MhqY KMjmSi59vZ	15-769-757-7250	22.71	AUTOMOBILE	ETHIOPIA	1998-06-15	579
8	42205	Customer#000042205	DgexH7dIC8FSbgO9p2tv	32-868-524-5077	469.60	MACHINERY	RUSSIA	1998-08-02	579
9	3451	Customer#000003451	UbM2A8GB2VFQ	15-160-369-7292	1720.73	MACHINERY	ETHIOPIA	1998-07-20	577
10	112753	Customer#000112753	CUxvi7gs,lN47k7	16-160-575-6122	3321.15	FURNITURE	FRANCE	1998-06-26	573

Figure 1: Top 10 Customers by RFM Score

Results (just now)

Table | Chart — 10 rows — 77ms

	ATION_NAME	LAST_ORDER_DATE	TOTAL_REVENUE	TOTAL_ORDERS	AVG_ORDER_VALUE	RECENCY_DAYS	R_SCORE	F_SCORE	M_SCORE	RFM_SCORE	RFM_SEGMENT
1	IL	1998-06-16	6294115.33	36	174836.54	47	5	5	5	15	Platinum (High Value)
2	OPIA	1998-08-02	6080943.83	37	164349.83	0	5	5	5	15	Platinum (High Value)
3	RIA	1998-07-08	6075141.96	37	164193.03	25	5	5	5	15	Platinum (High Value)
4	ED KINGDOM	1998-07-27	5848849.18	36	162468.03	6	5	5	5	15	Platinum (High Value)
5	A	1998-07-22	5842519.99	37	157905.95	11	5	5	5	15	Platinum (High Value)
6	CE	1998-08-02	5823782.70	32	181993.21	0	5	5	5	15	Platinum (High Value)
7	OPIA	1998-06-15	5798386.69	31	187044.73	48	5	5	5	15	Platinum (High Value)
8	IA	1998-08-02	5795404.01	36	160983.44	0	5	5	5	15	Platinum (High Value)
9	OPIA	1998-07-20	5777649.60	41	140918.28	13	5	5	5	15	Platinum (High Value)
10	CE	1998-06-26	5731193.61	37	154897.12	37	5	5	5	15	Platinum (High Value)

Figure 2: Top 10 Customers by RFM Score (continued)

Part 2 – Segment Exploration

Objective

To analyze the RFM segmentation results created in Part 1 and extract insights into customer distribution, revenue contribution, and key markets for high-value customers.

Tasks & Queries

All tasks below use the final RFM table (final_rfm) from Part 1.

For each task, replace the final SELECT * FROM final_rfm with the corresponding query to extract the required insights.

Task 1 - Number of Customers per RFM Segment

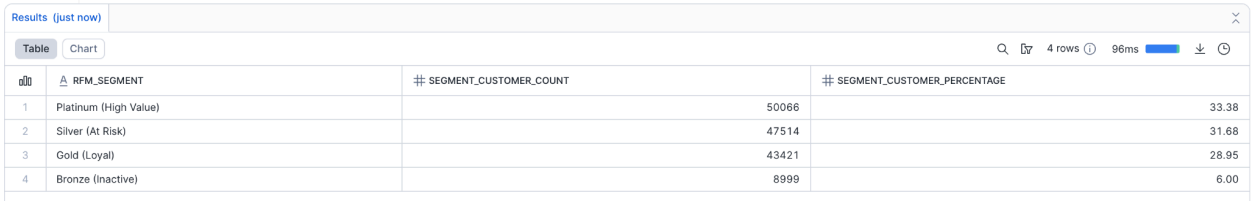
SQL Query

```sql
-- This query summarizes customers by RFM segment
-- and calculates the percentage of total customers
SELECT
    rfm_segment,

    -- Number of unique customers in each segment
    COUNT(DISTINCT customer_id) AS segment_customer_count,

    -- Percentage of total customers in each segment (rounded to 2
decimals)
    ROUND(
        COUNT(DISTINCT customer_id) * 100.0 / SUM(COUNT(DISTINCT
customer_id)) OVER (),
        2
    ) AS segment_customer_percentage
FROM
    final_rfm
GROUP BY
    rfm_segment
ORDER BY
    segment_customer_count DESC;
```

Output Screenshot



Results (just now)

Table | Chart — 4 rows — 96ms

	RFM_SEGMENT	SEGMENT_CUSTOMER_COUNT	SEGMENT_CUSTOMER_PERCENTAGE
1	Platinum (High Value)	50066	33.38
2	Silver (At Risk)	47514	31.68
3	Gold (Loyal)	43421	28.95
4	Bronze (Inactive)	8999	6.00

Figure 3: Number of Customers per RFM Segment

Key Insights

The Platinum segment has the most customers, showing strong engagement from a large part of the customer base. The Silver and Gold groups also include many customers, while the Bronze segment is much smaller, indicating fewer low-activity customers.

Task 2 - Total Revenue by RFM Segment

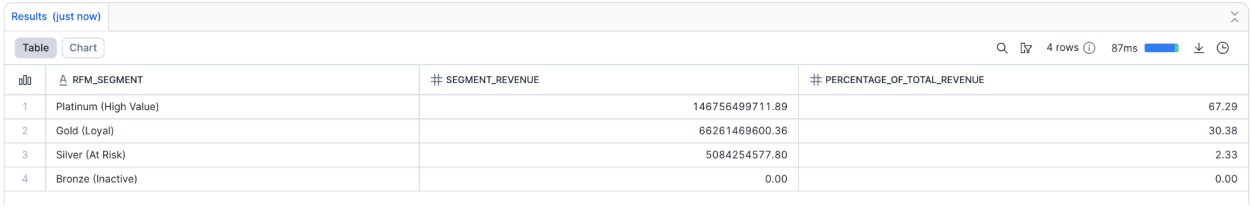
SQL Query

```sql
-- Total revenue contribution by each RFM segment
SELECT
    rfm_segment,

    -- Total revenue generated by the segment
    SUM(total_revenue) AS segment_revenue,

    -- Percentage of total segment revenue
    ROUND(
        SUM(total_revenue) * 100.0 / SUM(SUM(total_revenue)) OVER (),
        2
    ) AS percentage_of_total_revenue
FROM
    final_rfm
GROUP BY
    rfm_segment
ORDER BY
    segment_revenue DESC;
```

Output Screenshot



Results (just now)

Table | Chart — 4 rows — 87ms

	RFM_SEGMENT	SEGMENT_REVENUE	PERCENTAGE_OF_TOTAL_REVENUE
1	Platinum (High Value)	146756499711.89	67.29
2	Gold (Loyal)	66261469600.36	30.38
3	Silver (At Risk)	5084254577.80	2.33
4	Bronze (Inactive)	0.00	0.00

Figure 4: Total Revenue by RFM Segment

Key Insights

The Platinum segment generates most of the revenue, followed by Gold customers, while Silver contributes very little and Bronze generates none. This shows that top-tier customers drive nearly all of the company's sales.

Task 3 - Top 5 High-Value Customers by RFM Score

SQL Query

```sql
-- Top 5 Platinum (High Value) customers by RFM score and revenue
SELECT
    customer_id,       -- Unique customer identifier
    customer_name,     -- Customer name
    customer_address,  -- Customer address
    customer_phone,    -- Customer phone number
    nation_name,       -- Customer nation
    total_revenue,     -- Total revenue generated by the customer
    total_orders,      -- Total number of orders
    rfm_score,         -- Combined RFM score
    rfm_segment        -- Customer RFM segment
FROM
    final_rfm
WHERE
    rfm_segment = 'Platinum (High Value)' -- Filter to top-tier customers
ORDER BY
    rfm_score DESC,         -- Rank by RFM score
    total_revenue DESC      -- Tie-breaker by revenue
LIMIT
    5;
```

Output Screenshot

Results (just now)

Table | Chart — 5 rows — 80ms

	CUSTOMER_ID	CUSTOMER_NAME	CUSTOMER_ADDRESS	CUSTOMER_PHONE	NATION_NAME	TOTAL_REVENUE	TOTAL_ORDERS	RFM_SCORE	RFM_SEGMENT
1	95257	Customer#000095257	aYNvFHtnZ3nfM3KhZSS6alZbWfj407VfjEyQ8O	12-704-221-9665	BRAZIL	6294115.33	36	15	Platinum (High Value)
2	131113	Customer#000131113	Kfg65w3gklTu44xn15QOLy	15-698-563-4092	ETHIOPIA	6080943.83	37	15	Platinum (High Value)
3	134380	Customer#000134380	j0J3qLKeB6clpFxNDB46lDrZyYzT	10-184-851-8092	ALGERIA	6075141.96	37	15	Platinum (High Value)
4	20194	Customer#000020194	HWUhzU D4OeqKW5qrcvb7tKTH4N4RDE	33-506-798-6469	UNITED KINGDOM	5848849.18	36	15	Platinum (High Value)
5	148750	Customer#000148750	4DOJ6YXrSlGexdU,Hhhd1ET,	28-219-585-6544	CHINA	5842519.99	37	15	Platinum (High Value)

Figure 5: Top 5 High-Value Customers by RFM Score

Key Insights

The top five Platinum customers all have the maximum RFM score of 15 and generate over $5.8 million each from 36–37 orders, highlighting their high value and importance to the business.

Task 4 - Top Nations with High-Value Customers

SQL Query

```sql
-- Count and percentage of High Value (Platinum) customers by nation
-- Top 5 nations with the highest number of High Value (Platinum) customers
SELECT
    nation_name,

    -- Number of High Value customers
    COUNT(DISTINCT customer_id) AS high_value_customer_count,

    -- % of Total High Value customers
    ROUND(
        COUNT(DISTINCT customer_id) * 100.0 / SUM(COUNT(DISTINCT
customer_id)) OVER (),
        2
    ) AS high_value_customer_percentage
FROM
    final_rfm
WHERE
    rfm_segment = 'Platinum (High Value)'
GROUP BY
    nation_name
ORDER BY
    high_value_customer_count DESC -- Rank nations by High Value customers
LIMIT
    5;
```

Output Screenshot

Results (just now)

Table | Chart — 5 rows — 87ms

	NATION_NAME	HIGH_VALUE_CUSTOMER_COUNT	HIGH_VALUE_CUSTOMER_PERCENTAGE
1	INDONESIA	2095	4.18
2	FRANCE	2054	4.10
3	JORDAN	2040	4.07
4	CANADA	2040	4.07
5	BRAZIL	2033	4.06

Figure 6: Top Nations with High-Value Customers

Key Insights

Indonesia, France, Jordan, Canada, and Brazil have the highest number of High Value customers, each contributing around 4% of the segment. This indicates these nations are key markets for targeting and retaining top-tier customers.

Part 3 – Written Insights

1. RFM Scoring and Segmentation Approach

I used the **RFM (Recency, Frequency, Monetary)** method to measure how valuable and engaged each customer is. Each customer got a score from **1 to 5** for Recency, Frequency, and Monetary, where **5** means very recent purchases, frequent orders, or high spending. For customers with no orders, I used a default date (**'1900-01-01'**) to mark them as **inactive**. I then added the three scores to get a **total RFM score out of 15**. Based on this score, customers were grouped into four segments: **Platinum (12–15), Gold (8–11), Silver (4–7), and Bronze (below 4)**. This helps identify **top customers** and those who may need more attention.

2. Key RFM Driver

From the analysis, **monetary value** is the biggest factor driving customer behavior. The **top 10 customers** all spend over **5.7 million**, which strongly determines their **RFM ranking**. Recency (**0–48 days**) and Total Orders (**31–41**) vary, but **revenue** is what really separates the top customers. This shows that **high spenders are the most important**, and marketing should focus on keeping these customers engaged. **Recency and Frequency** are still useful, but secondary.

3. Segment Distribution Insights

The customer base is distributed as follows: **Platinum 33.4%, Silver 31.7%, Gold 28.9%, Bronze 6%**. Most customers are **mid-tier**, either moderately engaged or at risk of churning. **Platinum customers**, even though only a third of the base, generate the **most revenue**. The small **Bronze group** shows low inactivity, which is good, but there is still a chance to **re-engage mid-tier customers**. Overall, the company should focus on **keeping Platinum customers happy** while increasing engagement in **Silver and Gold segments**.

4. Strategic Recommendations

For **Platinum customers**, the company should provide **personalized rewards, VIP programs, or exclusive offers** to maintain loyalty. **Gold and Silver customers** can be targeted with campaigns to encourage **more frequent orders, upselling, or cross-selling**. **Bronze customers** may need **incentives or reactivation campaigns** to prevent churn. **Regional patterns**, like top customers concentrated in certain countries, can guide **localized marketing**. Overall, the focus should be on **retaining high-value customers** and improving **mid-tier engagement** to maximize revenue.

5. Further Analysis

To understand customer behavior better, additional analyses can help. **Lifecycle analysis** can show where customers are in their journey and highlight **at-risk or inactive groups**. **CLV (Customer Lifetime Value) and CLV bands** can identify the most valuable customers and guide **resource allocation**. **Retention factor** can predict who is likely to buy again, helping prevent churn. **Cohort and purchase behavior analysis** can reveal trends in **frequency or product preference over time**. Combining **RFM with geography and demographics** can help create **personalized and local marketing campaigns**.